

independently and primarily at the beginning of a text sequence.” Lin et al. [25] also find that differences between aligned and unaligned base models introduced by the alignment fine-tuning vanishes as the sequence goes longer (similar to the effect that we observe in Figure 1, though theirs are not in safety-specific contexts). Particularly, while Lin et al. [25] primarily tie such effects to question whether fine-tuning is even necessary to achieve current level of alignment (e.g., in-context learning may already suffice to achieve a comparable level of alignment), we go much deeper into investigating the safety-specific effects of this phenomenon and tie it to multiple downstream attacks and training-based mitigations. Besides, we find Zhao et al. [27] also note a similar token-wise effect, and they explicitly exploit this effect to design jailbreak attacks. Others have investigated fine-tuning dynamics through interpretability or pruning methods, which is distinct but somewhat related to the approach we take here [51, 52].

Protecting The Safety Alignment at Initial Token Positions. In Section 4, one important insight that motivates the design of our constrained fine-tuning loss is that “safety alignment would be more difficult to be circumvented if we can protect the generative distribution of the model at the early token positions.” We note that Xu et al. [53] share a similar insight to ours in this regard. They find it is possible to design a defense against inference-time jailbreak attacks by simply identifying safety disclaimers and amplifying their token probabilities at the early token positions.

Connections to Control Theory and Safe Reinforcement Learning. Our data augmentation approach in Section 3 relates to exploration requirements for optimal learning via policy gradient methods [54], learning recovery policies [55], and safe control theory [56, 57]. We, however, leave deeper connections to this literature for future work.

Other Notions of Safety Depth. We also note that safety “depth” may be multi-dimensional in addition to token-based depth we describe here. Other considerations for depth would be the ability for models to retain safety properties after adaptation that some have previously discussed [58, 51, 22].

6 Conclusion

Our work identifies a shortcut that current safety alignment strategies appear to exploit: that alignment only needs to change the generative distribution of the first few tokens. We show that this may be a key component of many downstream vulnerabilities. We then provide two initial strategies to address this: (1) a data augmentation approach that can increase depth of alignment; (2) a constrained optimization objective that can help mitigate finetuning attacks by constraining updates on initial tokens. Future work can explore additional approaches grounded in control theory and safe reinforcement learning. The methods we describe here may not be a perfect defense and may be subject to some future adaptive attacks, but they are an initial step for improving robustness and further demonstrate how much improvement there can be over current approaches. Fundamentally, our results are primarily to support our argument that future safety alignment should be made more than just a few tokens deep.

Broader Impacts and Ethics Statement

Our work explicitly ties failure modes of safety alignment to potential shortcuts that can be taken by alignment methods and advocates for, and provides a path forward for, deeper alignment approaches that will improve safety more broadly. While a deeper understanding of the failures of alignment may result in increased ability to jailbreak models, we believe that open investigations of such failure modes are important for strengthening the safety of future models and broadly ensuring that models have positive societal impact. The proposed prototype approaches (in this work) for strengthening the alignment in current LLMs also contribute to the broader agenda of building safer and secure AI.

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